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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 12/29/2023 |
| **PI Name:** | Matson, Douglas Ph.D. | | |
| **Project Title:** | Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | douglas.matson@tufts.edu | **Fax:** | FY (617) 627-3058 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (617) 627-3239 |
| **Organization Name:** | Tufts University | | |
| **PI Address 1:** | Mechanical Engineering | | |
| **PI Address 2:** | 025 Anderson Hall, 200 College Ave | | |
| **PI Web Page:** | https:// | | |
| **City:** | Medford | **State:** | MA |
| **Zip Code:** | 02155 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms |
| **Start Date:** | 12/01/2015 | **End Date:** | 02/29/2024 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Sansoucie, Michael | **Contact Phone:** | 256.544.5269 |
| **Contact Email:** | michael.p.sansoucie@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19)<br>NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC19K0256 ; NNX16AB59G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC).<br>Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports.<br><br>Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted.<br><br>Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany<br><br>Subproject: Transformation kinetics - ISS (International Space Station)<br><br>Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.<br><br>Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany<br><br>Subprojects:<br><br>Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies.<br><br>Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys.<br><br>Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity.<br><br>Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS.<br><br>MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter. |
| **Task Progress:** | Highlights of Work Accomplished During Grant Period Mar 2023 - Feb 2024:<br>[Ed. Note: For additional information and full citations, see the Cumulative Bibliography for this project.]<br><br>• Graduated 2 MS and 1 BS senior honors thesis student.<br><br>• Published 9 peer-reviewed articles, 3 thesis submissions, and 10 invited talks.<br><br>• Attended 39th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR-39) 2023 Conference, Washington DC, November 15-18 as organizer and as lead for the International Space Station National Laboratory (ISSNL) User Advisory Committee Public Meeting.<br><br>• Developed test plan documentation to run future Batch 3/4 alloys using the International Space Sation (ISS) Electromagnetic Levitator (EML). (FeCo30, FeNi10, HEA, FeCB, TiZrNi). Developed test plan and delivered ground support and flight samples for Batch 4 (Inconel 626, Inconel 718, and Gold). Supported proposal collaborations for Batch 5 candidates.<br><br>• Honored as 2023 NASA International Space Station Research and Development Conference (ISSRDC) Award for Compelling Results in Physical Sciences & Materials.<br><br>Highlights of Previous Work Accomplished:<br><br>• Implemented Batch 1, Batch 2, and Batch 3 operations using ISS-EML on orbit. (Fe-20Cr-20Ni, Fe-16Cr-12Ni, Fe-12Ni-16Cr, FeCo50, FeCo55, FeCo40 TiZrNi, FeB, NdFeB, Ti48Al48Cr2Nb2, CMSX-10, LEK94, MC2, FeCrNiCo HEI, CoSi62).<br><br>• Conducted parabolic flight and ground test experiment plan to support Batch 4 alloys. (Au, CMSX-4+, EHEA, superalloys In625/In718, FeCo30).<br><br>• Investigated thermophysical property measurement for control of convection using the NASA Marshall Space Flight Center (MSFC) electrostatic levitator (ESL), the portable EML at the German Synchrotron (EML-DESY), and Korea Research Institute of Standards and Science (KRISS) ESL in support of Batches 1-4.<br><br>• Ran tests at DESY German Synchrotron to define transformation pathway for alloy rapid solidification using Dresden EML facility and at the Advanced Photon Source (APS) at Argonne National Laboratory, using Tufts Glass-flux |

technique.

• Served as European Space Agency (ESA) Investigator Working Group (IWG) chair and NASA ISS-EML facility scientist over the entire program period. • Developed a new theoretical model to explain metastable transformations during rapid solidification of steel alloys to include the influence of melt convection with application to the welding and continuous casting industry.

• Developed a novel interpretation of how dislocation density plays an important role in defining how heterogeneous nucleation occurs along pre-existing grain boundaries in steel; impacts how investment casting systems are designed.

• Developed a new approach to understanding the influence of deformation on the accuracy and precision of surface tension measurements for superalloys important for NASA exploration mission success.

• Featured speaker for NASA International Space Station Research and Development (ISSR&D), American Society for Gravitational and Space Research (ASGSR), International Conference on Advances in Solidification Processes (ICASP5), Japanese Conference on Microgravity Science and Application (JASMAC-31, JASMAC-32 and JASMAC-33 conferences).

• Honored as German Materials Society (DGM) Köster Award 2021 ( <a target="_blank" href="https://doi.org/10.1515/ijmr-2021-8266">https://</a> ).

• Recognized as a Fellow for ASGSR space science professional society.

• Recognized as a Fellow for ASM International materials science professional society. [Ed. Note: ASM International, formerly known as the American Society for Metals, is an association for materials science engineers and scientists.]

• Served as Chair of the ISS-NL User Advisory Committee supporting Center for the Advancement of Science in Space (CASIS) operations (Ops).

• Served as Co-Chair of the National Academies Committee on Biological and Physical Sciences in Space to oversee the 2023 NASA Decadal Study.

| Bibliography Type: | Description: (Last Updated: 05/05/2026) |
|---|---|
| Articles in Peer-reviewed Journals | Matson DM, Battezzati L, Galenko PK, Gandin CA, Gangopadhyay AK, Henein H, Kelton KF, Kolbe M, Valloton J, Vogel SC, Volkmann T. "Electromagnetic levitation containerless processing of metallic materials in microgravity: Rapid solidification." npj Microgravity. 2023 Aug 15;9(1):65. https://www.nature.com/articles/s41526-023-00310-2 , Aug-2023 |
| Articles in Peer-reviewed Journals | Nawer J, Ishikawa T, Oda H, Koyama C, Saruwatari H, Sansoucie MP, Philips B, Gossé S, Mohr M, Kolbe M, Matson DM. "A quantitative comparison of thermophysical property measurement of CMSX-4® Plus (SLS) in microgravity and terrestrial environments." High Temperatures-High Pressures. 2023 May 1;52:323-39. http://dx.doi.org/10.32908/hthp.v52.1407 , May-2023 |
| Articles in Peer-reviewed Journals | Nawer J, Ishikawa T, Oda H, Saruwatari H, Koyama C, Xiao X, Schneider S, Kolbe M, Matson DM. "Uncertainty analysis and performance evaluation of thermophysical property measurement of liquid Au in microgravity." npj Microgravity. 2023 May 24;9(1):38. https://doi.org/10.1038/s41526-023-00277-0 , May-2023 |
| Articles in Peer-reviewed Journals | Mohr M, Dong Y, Bracker GP, Hyers RW, Matson DM, Zboray R, Frison R, Dommann A, Neels A, Xiao X, Brillo J. "Electromagnetic levitation containerless processing of metallic materials in microgravity: Thermophysical properties." npj Microgravity. 2023 May 2;9(1):34. https://doi.org/10.1038/s41526-023-00281-4 , May-2023 |
| Articles in Peer-reviewed Journals | Nawer J, Matson DM. "Quantifying facility performance during thermophysical property measurement of liquid Zr using Electrostatic Levitation." High Temperatures-High Pressures. 2023 Mar 1;52(2):123-38. http://dx.doi.org/10.32908/hthp.v52.1315 , Mar-2023 |
| Articles in Peer-reviewed Journals | Azersky C, Jeon S, Settens C, Fakhrul M, Matson D. "Quantifying X-ray diffractometer precision for measurement of dislocation density and the evolution of defect structures during solidification of additive manufacturing powders." IOP Conference Series: Materials Science and Engineering. 2023 May 1;1281(1):012033. IOP Publishing. http://dx.doi.org/10.1088/1757-899X/1281/1/012033 , May-2023 |
| Articles in Peer-reviewed Journals | Nawer J, Ishikawa T, Oda H, Koyama C, Matson DM. "Uncertainty quantification of thermophysical property measurement in space and on Earth: A study of liquid Platinum using electrostatic levitation." J Astron Space Sci. 2023 Sep 15;40(3):93-100. https://doi.org/10.5140/JASS.2023.40.3.93 , Sep-2023 |
| Articles in Peer-reviewed Journals | Sillekens WH, Sansoucie MP, Hyers RW, Bracker GP, Matson DM. "Forward: Materials research in reduced gravity 2023." Metall Mater Trans A 54. 2023 Nov;54(11):4139-40. https://doi.org/10.1007/s11661-023-07195-6 , Nov-2023 |
| Articles in Peer-reviewed Journals | Muusha P, Stanford B, Kolbe M, Matson DM. "Relating cooling rates in superheated liquid and during solidification for powder characterization." Metall Mater Trans A 54. 2023 Nov;54(11):4169-78. https://doi.org/10.1007/s11661-023-07135-4 , Nov-2023 |
| Articles in Peer-reviewed Journals | Nawer J, Stanford B, Kolbe M, Schneider S, Gossé S, Wunderlich RK, Mohr M, Borzì A, Neels A, Matson DM. "Thermodynamic assessment of evaporation during molten steel testing onboard the International Space Station." npj Microgravity. 2024 Jul 19;10(1):77. https://doi.org/10.1038/s41526-024-00416-1 , Jul-2024 |
| Articles in Peer-reviewed Journals | Jeon S, Lee JJ, Kaban I, Matson DM. "In situ observation of a transient intermediate state during metastable solidification of a Fe–Cr–Ni alloy." Acta Materialia. 2026 Apr 15;308:122028. https://doi.org/10.1016/j.actamat.2026.122028 , Apr-2026 |
| Awards | Moldovean N. "Matthew Isakowitz Fellowship (internship with Redwire)." Dec-2023 |

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| **Awards** | Matson DM. "International Space Station Research and Development Conference (ISSRDC) Award for Compelling Results in Physical Sciences & Materials." Aug-2023 |
| **Dissertations and Theses** | Waskow D. "The effect of undercooling in an Fe-Ni alloy." BS Thesis, Tufts University Mechanical Engineering Department, February 2023. , Feb-2023 |
| **Dissertations and Theses** | Muusha P. "Relating cooling rates in superheated liquid and during solidification for powder characterization." MS Thesis, Tufts University Mechanical Engineering Department, February 2024. , Feb-2024 |
| **Dissertations and Theses** | Stanford B. "A saturation approach to delay time estimation during the solidification of ferrous alloys." MS Thesis, Tufts University Mechanical Engineering Department, February 2024. , Feb-2024 |